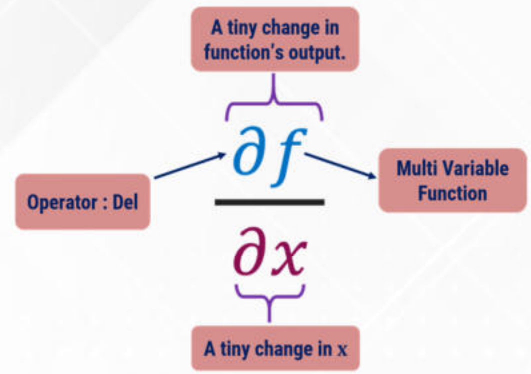
Mathematics-I (M-I)
GTU # 3110014

Unit-4
Partial Derivatives



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Method:7 → Local Extreme Values

Local Extreme Values of Function of Two Variables

- A point (a, b) is said to be a stationary point of $f(x, y)$ if

$$\left(\frac{\partial f}{\partial x}\right)_{(a,b)} = 0 \quad \& \quad \left(\frac{\partial f}{\partial y}\right)_{(a,b)} = 0.$$

- A stationary point (a, b) is said to be a saddle point of $f(x, y)$ if at point (a, b) given function has neither local minima nor local maxima.
- A local minimum OR local maximum of a function is called local extreme values.

Procedure to Find Local Extreme Values of Function f(x, y)

(1). Find $\dfrac{\partial f}{\partial x}$ & $\dfrac{\partial f}{\partial y}$.

(2). Solve $\dfrac{\partial f}{\partial x}=0, \dfrac{\partial f}{\partial y}=0$ to find the stationary points $(a_1, b_1), (a_2, b_2)$, ...

(3). Find $r=\dfrac{\partial^2 f}{\partial x^2}, s=\dfrac{\partial^2 f}{\partial x \partial y}$ & $t=\dfrac{\partial^2 f}{\partial y^2}$.

(4). Find $r=\left(\dfrac{\partial^2 f}{\partial x^2}\right)_{(a_1,b_1)}, s=\left(\dfrac{\partial^2 f}{\partial x \partial y}\right)_{(a_1,b_1)}$ & $t=\left(\dfrac{\partial^2 f}{\partial y^2}\right)_{(a_1,b_1)}$.

Procedure to Find Local Extreme Values of Function f(x, y)

$(\mathbf{5})$. Find $rt - s^2$ for stationary point (a_1, b_1).

$(\mathbf{6})$. Conclusion for stationary point (a_1, b_1) from the following table.

RESULT	CONCLUSION
$rt - s^2 > 0 \;\&\; r > 0$	Function has local minimum value
$rt - s^2 > 0 \;\&\; r < 0$	Function has local maximum value
$rt - s^2 < 0$	Function has no local maxima or local minima (i.e. saddle point)
$rt - s^2 = 0 \quad \text{OR} \quad r = 0$	Nothing can be said about the local maxima or local minima. It requires further investigation.

Procedure to Find Local Extreme Values of Function f(x, y)

$(\mathbf{7})$. Repeat step (4), (5) & (6) for all stationary points.

$(\mathbf{8})$. Substitute the point (a_i, b_i) in the given function to find the local maximum or local minimum value of the function at that point, if possible.

Example of Local Extreme Values

Example 2: Find extreme values of $f(x,y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$.

Solution: Here, $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$.

$$\frac{\partial f}{\partial x} = 3x^2 + 3y^2 - 30x + 72 \qquad \frac{\partial f}{\partial y} = 6xy - 30y$$

$$r = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial x}(\qquad) = 6x - 30$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial x}(\qquad) = 6y$$

$$t = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial y}(\qquad) = 6x - 30$$

Example of Local Extreme Values

Solution Continue: For stationary point $\frac{\partial f}{\partial x} = 0$ & $\frac{\partial f}{\partial y} = 0.$

$$3x^2 + 3y^2 - 30x + 72 = 0$$
$$\Rightarrow 3(x^2 + y^2 - 10x + 24) = 0$$
$$\Rightarrow \boxed{x^2 + y^2 - 10x + 24 = 0}$$

$$6xy - 30y = 0$$
$$\Rightarrow 6y(x - 5) = 0$$
$$\Rightarrow y(x - 5) = 0$$
$$\Rightarrow y = 0 \quad \text{OR} \quad x - 5 = 0$$
$$\Rightarrow \boxed{y = 0} \quad \text{OR} \quad \boxed{x = 5}$$

Case: 1 $\quad y = 0$

$$\Rightarrow x^2 + y^2 - 10x + 24 = 0$$
$$\Rightarrow x^2 + (\quad)^2 - 10x + 24 = 0$$
$$\Rightarrow x^2 - 10x + 24 = 0$$
$$\Rightarrow x^2 - 4x - 6x + 24 = 0$$
$$\Rightarrow x(x - 4) - 6(x - 4) = 0$$
$$\Rightarrow x - 4 = 0 \quad \text{OR} \quad x - 6 = 0$$
$$\Rightarrow x = 4 \quad \text{OR} \quad x = 6$$

Example of Local Extreme Values

Solution Continue:

Case: 2 $x = 5$

$\Rightarrow x^2 + y^2 - 10x + 24 = 0$

$\Rightarrow (\quad)^2 + y^2 - 10(\quad) + 24 = 0$

$\Rightarrow 25 + y^2 - 50 + 24 = 0$

$\Rightarrow y^2 - 1 = 0$

$\Rightarrow y^2 = 1$

$\Rightarrow y = 1$ OR $y = -1$

Case: 1 $y = 0$

$\Rightarrow x = 4$ OR $x = 6$

Hence, stationary points are

$(\ ,\ )$ & $(\ ,\ )$.

Case: 2 $x = 5$

$\Rightarrow y = 1$ OR $y = -1$

Hence, stationary points are

$(\ ,\ )$ & $(\ ,\ )$.

Example of Local Extreme Values

Solution Continue:

Stationary Points	$r = 6x - 30$	$s = 6y$	$t = 6x - 30$	$rt - s^2$	Conclusion
$(4, 0)$	-6	0	-6	$36 > 0 (r < 0)$	Local Maxima
$(6, 0)$	6	0	6	$36 > 0 (r > 0)$	Local Minima
$(5, 1)$	0	6	0	$-36 < 0$	Saddle Point
$(5, -1)$	0	-6	0	$-36 < 0$	Saddle Point

Example of Local Extreme Values

Example 2: Find extreme values of $f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$.

Solution Continue:

$$\operatorname{Max}(f) = f(x, y)_{(4,0)} = (4)^3 + 3(4)(0)^2 - 15(4)^2 - 15(0)^2 + 72(4)$$

$$= 4^2(4 - 15 + 18) = 16(7) = 112$$

$$\Rightarrow \boxed{\operatorname{Max}(f) = 112}$$

$$\operatorname{Min}(f) = f(x, y)_{(6,0)} = (6)^3 + 3(6)(0)^2 - 15(6)^2 - 15(0)^2 + 72(6)$$

$$= 6^2(6 - 15 + 12) = 36(3) = 108$$

$$\Rightarrow \boxed{\operatorname{Min}(f) = 108}$$

Example of Local Extreme Values

Example 1: Find the local extreme values of $f(x,y) = x^2 - y^2 - xy - x + y + 6$, if possible.

Solution: Here, $f(x,y) = x^2 - y^2 - xy - x + y + 6$.

$$\frac{\partial f}{\partial x} = 2x - y - 1$$

$$\frac{\partial f}{\partial y} = -2y - x + 1$$

$$r = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial x}(\qquad) = 2$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial x}(\qquad) = -1$$

$$t = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial y}(\qquad) = -2$$

Example of Local Extreme Values

Solution Continue: For stationary point $\frac{\partial f}{\partial x} = 0$ & $\frac{\partial f}{\partial y} = 0.$

$$\Rightarrow 2x - y - 1 = 0 \quad \& \quad -2y - x + 1 = 0$$

$$\Rightarrow \quad 2x - y = 1 \qquad (1)$$

$$\& \quad x + 2y = 1 \qquad (2)$$

$$\Rightarrow \begin{array}{rl} 4x - \cancel{2y} = 2 & 2 \times eq^n(1) \\ +\quad x + \cancel{2y} = 1 & (2) \\ \hline 5x \qquad = 3 & \end{array}$$

$$\Rightarrow \boxed{x = \frac{3}{5}}$$

$$2x - y = 1$$

$$\Rightarrow y = 2x - 1$$

$$= 2\left(\quad \right) - 1$$

$$= \left(\frac{6}{5}\right) - 1$$

$$\Rightarrow \boxed{y = \frac{1}{5}}$$

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Example of Local Extreme Values

Solution Continue:

Hence, stationary point is $\left(\frac{3}{5}, \frac{1}{5}\right)$.

Here, $r = 2, s = -1 \;\&\; t = -2$.

$\Rightarrow r = 2, s = -1 \;\&\; t = -2$ at $\left(\frac{3}{5}, \frac{1}{5}\right)$

$\Rightarrow rt - s^2 = (2)(-2) - (-1)^2 = -4 - 1 = -5$

$\Rightarrow rt - s^2 < 0$

$\Rightarrow \boxed{\left(\frac{3}{5}, \frac{1}{5}\right) \text{ is saddle point}}.$

Example of Local Extreme Values

Example 8: For what values of the constant k does the second derivative test guarantee that $f(x,y) = x^2 + kxy + y^2$ will have a saddle point at $(0,0)$?A local minimum at $(0,0)$?

Solution: Here, $f(x,y) = x^2 + kxy + y^2$.

$$\frac{\partial f}{\partial x} = 2x + ky$$

$$\frac{\partial f}{\partial y} = kx + 2y$$

$$r = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial x}(\qquad) = 2$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial x}(\qquad) = k$$

$$t = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial y}(\qquad) = 2$$

Example of Local Extreme Values

Solution Continue: For stationary point $\frac{\partial f}{\partial x} = 0 \;\&\; \frac{\partial f}{\partial y} = 0.$

$$\Rightarrow \quad 2x + ky = 0$$

$$\& \quad kx + 2y = 0$$

$$\Rightarrow \begin{array}{r} 4x \quad + \cancel{2ky} = 0 \\ -k^2x - \cancel{2ky} = 0 \\ \hline (4 - k^2)x = 0 \end{array}$$

$$\Rightarrow \boxed{x = 0}$$

$$2x + ky = 0$$

$$\Rightarrow 2(\quad) + ky = 0$$

$$\Rightarrow ky = 0$$

$$\Rightarrow \boxed{y = 0}$$

Hence, stationary point is $(0, 0)$.

Example of Local Extreme Values

Solution Continue:

- For saddle point at $(0, 0)$,

  $(rt - s^2)_{(0,0)} < 0.$

  $\Rightarrow (2)(2) - k^2 < 0$

  $\Rightarrow 4 - k^2 < 0 \;\Rightarrow k^2 > 4$

  $\Rightarrow \boxed{k \in \mathbb{R} - [-2, 2]}$

- For local minimum at $(0, 0)$,

  $(rt - s^2)_{(0,0)} > 0, (r > 0/t > 0).$

  $\Rightarrow (2)(2) - k^2 > 0$

  $\Rightarrow 4 - k^2 > 0 \;\Rightarrow k^2 < 4$

  $\Rightarrow \boxed{k \in (-2, 2)}$

Method:8 → Lagrange's Multiplier

Lagrange's Multiplier Method

- This method is apply for finding maxima & minima of given function with satisfying given condition(constraint).
- The drawback of this method is that it gives no information about the nature of stationary point directly i.e. about maxima and minima.
- Here unknown constant λ is known as Lagrange's multiplier.
- The distance between two points $P(x, y, z)$ & $Q(a, b, c)$ is $PQ = \sqrt{(x-a)^2 + (y-b)^2 + (z-c)^2}$.

Lagrange's Multiplier Method

Procedure for find finding maxima & minima of the function $f(x, y)$ with satisfying the condition $\phi(x, y) = 0$ is as follow:

$(\mathbf{1})$. Construct the new function $F(x, y)$ as $F(x, y) = f(x, y) + \lambda\,\phi(x, y)$.

$(\mathbf{2})$. Solve the three equations $\dfrac{\partial F}{\partial x} = 0, \dfrac{\partial F}{\partial y} = 0$ & $\phi(x, y) = 0$ to find the stationary points $(x_1, y_1), (x_2, y_2), \ldots$.

$(\mathbf{3})$. Find the value of $f(x, y)$ at above stationary points and give the conclusion about maxima or minima.

Example of Lagrange's Multiplier Method

Example 2: Find the maximum and minimum values of function $f(x,y) = 3x + 4y$ on the circle $x^2 + y^2 = 1$ using the method of Lagrange multiplier.

Solution: Here, $f(x,y) = 3x + 4y$ and $\phi(x,y) = x^2 + y^2 - 1 = 0$.

- Construct the new function $F(x,y)$ as $F(x,y) = f(x,y) + \lambda\,\phi(x,y)$.

$\Rightarrow F(x,y) = (\qquad\qquad) + \lambda(\qquad\qquad)$

- Solve three equations $\dfrac{\partial F}{\partial x} = 0, \dfrac{\partial F}{\partial y} = 0$ & $\phi(x,y) = 0$ for stationary points.

$$\frac{\partial F}{\partial x} = 3 + \lambda(2x) = 0$$

$$\Rightarrow \boxed{\lambda = -\frac{3}{2x}} \dashrightarrow (1)$$

$$\frac{\partial F}{\partial y} = 4 + \lambda(2y) = 0$$

$$\Rightarrow \boxed{\lambda = -\frac{4}{2y}} \dashrightarrow (2)$$

$$x^2 + y^2 - 1 = 0$$

$$\Rightarrow \boxed{x^2 + y^2 = 1} \dashrightarrow (3)$$

Example of Lagrange's Multiplier Method

$$\lambda = -\frac{3}{2x} \quad --\rightarrow \mathbf{(1)}$$

$$\lambda = -\frac{4}{2y} \quad --\rightarrow \mathbf{(2)}$$

$$x^2 + y^2 = 1 \quad --\rightarrow \mathbf{(3)}$$

Solution Continue:

- From equations (1) & (2)

$$\cancel{-}\frac{3}{\cancel{2}x} = \cancel{-}\frac{4}{\cancel{2}y} \Rightarrow \boxed{y = \frac{4x}{3}}$$

Substitute in (3)

$$\Rightarrow x^2 + \left(\frac{4x}{3}\right)^2 = 1$$

$$\Rightarrow x^2 + \frac{16x^2}{9} = 1$$

$$\Rightarrow 9x^2 + 16x^2 = 9$$

$$\Rightarrow 25x^2 = 9$$

$$\Rightarrow x^2 = \frac{9}{25}$$

$$\Rightarrow \boxed{x = \pm\frac{3}{5}}$$

- $y = \frac{4x}{3}$

$$= \frac{4}{\cancel{3}}\left(\pm\frac{\cancel{3}}{5}\right)$$

$$\Rightarrow \boxed{y = \pm\frac{4}{5}}$$

Example of Lagrange's Multiplier Method

Example 2: Find the maximum and minimum values of function $f(x,y) = 3x + 4y$ on the circle $x^2 + y^2 = 1$ using the method of Lagrange multiplier.

Solution Continue:

Hence, stationary points are $\left(\frac{3}{5}, \frac{4}{5}\right)$ & $\left(-\frac{3}{5}, -\frac{4}{5}\right)$.

$$f\left(\frac{3}{5}, \frac{4}{5}\right) = 3\left(\frac{3}{5}\right) + 4\left(\frac{4}{5}\right)$$

$$= \frac{9+16}{5} = \frac{25}{5}$$

$$\Rightarrow \boxed{f\left(\frac{3}{5}, \frac{4}{5}\right) = 5}.$$

Which is maximum value of $f(x,y)$.

Similarly, we get $\boxed{f\left(-\frac{3}{5}, -\frac{4}{5}\right) = -5}$.

Which is minimum value of $f(x,y)$.

Lagrange's Multiplier Method

- Procedure for find finding maxima & minima of the function $f(x, y, z)$ with satisfying the condition $\phi(x, y, z) = 0$ is as follow:

(1). Construct the new function $F(x, y, z)$ as $F(x, y, z) = f(x, y, z) + \lambda\,\phi(x, y, z)$.

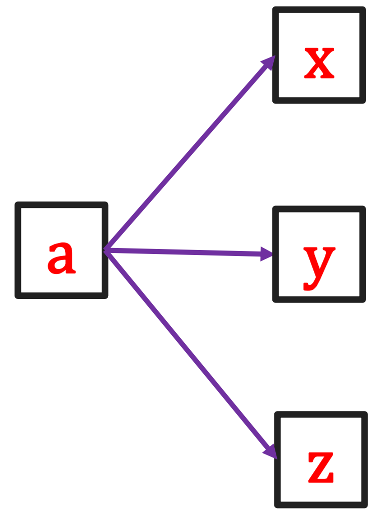
(2). Solve the four equations $\frac{\partial F}{\partial x} = 0, \frac{\partial F}{\partial y} = 0, \frac{\partial F}{\partial z} = 0$ & $\phi(x, y, z) = 0$ to find the stationary points $(x_1, y_1, z_1), (x_2, y_2, z_2), \dots$.

(3). Find the value of $f(x, y, z)$ at above stationary points and give the conclusion about maxima or minima.

Example of Lagrange's Multiplier Method

Example 5: Divided a given number 'a' into three positive points such that their sum is 'a' and product is maximum.

Solution: ➡ $x + y + z = a \;\; \& \; x \cdot y \cdot z$ is maximum.



Hence, $f(x, y, z) = xyz \;\; \& \; \phi(x, y, z) = x + y + z - a = 0.$

➡ Construct the new function $F(x, y, z)$ as

$$F(x, y, z) = f(x, y, z) + \lambda\, \phi(x, y, z).$$

$$\Rightarrow F(x, y, z) = (\qquad) + \lambda(\qquad\qquad\qquad)$$

➡ Solve the four equations $\dfrac{\partial F}{\partial x} = 0, \dfrac{\partial F}{\partial y} = 0, \dfrac{\partial F}{\partial z} = 0 \;\&\; \phi(x, y, z) = 0$ to find the stationary points.

Example of Lagrange's Multiplier Method

Solution Continue: $F(x, y, z) = (xyz) + \lambda(x + y + z - a)$

$$\frac{\partial F}{\partial x} = yz + \lambda(1) = 0$$

$$\Rightarrow \boxed{\lambda = -yz} \dashrightarrow (1)$$

$$\frac{\partial F}{\partial y} = xz + \lambda(1) = 0$$

$$\Rightarrow \boxed{\lambda = -xz} \dashrightarrow (2)$$

$$\frac{\partial F}{\partial z} = xy + \lambda(1) = 0$$

$$\Rightarrow \boxed{\lambda = -xy} \dashrightarrow (3)$$

From equations (1), (2) & (3) we get

$$-yz = -xz = -xy \quad \Rightarrow \boxed{yz = xz = xy}$$

$$yz = xz$$

$$\Rightarrow \boxed{y = x}$$

$$xz = xy$$

$$\Rightarrow \boxed{z = y}$$

$$x + y + z - a = 0$$

$$\Rightarrow \boxed{x + y + z = a} \dashrightarrow (4)$$

Example of Lagrange's Multiplier Method

Solution Continue:

$$x + y + z = a \dashrightarrow (4) \quad y = x \ \& \ z = y$$

$$\Rightarrow y + y + y = a$$

$$\Rightarrow 3y = a$$

$$\Rightarrow y = \frac{a}{3} \quad \Rightarrow x = \frac{a}{3} \ \& \ z = \frac{a}{3}$$

Hence, stationary point is $(x, y, z) = \left(\frac{a}{3}, \frac{a}{3}, \frac{a}{3}\right)$.

$$\Rightarrow f(x, y, z) = xyz = \frac{a}{3} \cdot \frac{a}{3} \cdot \frac{a}{3} = \frac{a^3}{27}$$

$$\Rightarrow \boxed{f(x, y, z) = \frac{a^3}{27}}$$

Which is maximum value of $f(x, y, z)$.

Example of Lagrange's Multiplier Method

Example 6: Find the point on the plane $x + 2y + 3z = 13$ closest to point $(1, 1, 1)$.

Solution: Let, $P(x, y, z)$ be the point on the plane $x + 2y + 3z = 13$ closest to the point $Q(1, 1, 1)$.

Hence distance between P & Q is minimum and which is defined as:

$$PQ = \sqrt{(x-1)^2+(y-1)^2+(z-1)^2}$$

$$\Rightarrow PQ^2 = (x-1)^2+(y-1)^2+(z-1)^2 \ \& \ x+2y+3z = 13$$

$$\Rightarrow f(x, y, z) = (x-1)^2+(y-1)^2+(z-1)^2$$

$$\& \ \phi(x, y, z) = x+2y+3z-13 = 0$$

Example of Lagrange's Multiplier Method

Solution Continue:

$f(x,y,z) = (x-1)^2 + (y-1)^2 + (z-1)^2$

$\phi(x,y,z) = x + 2y + 3z - 13 = 0$

- Construct the new function $F(x,y,z)$ as $F(x,y,z) = f(x,y,z) + \lambda\,\phi(x,y,z)$.

$\Rightarrow F(x,y,z) = (\qquad) + \lambda(\qquad)$

- Solve the four equations $\frac{\partial F}{\partial x} = 0, \frac{\partial F}{\partial y} = 0, \frac{\partial F}{\partial z} = 0$ & $\phi(x,y,z) = 0$ to find the stationary points.

- $\frac{\partial F}{\partial x} = 2(x-1) + \lambda(1) = 0$

$\Rightarrow \boxed{\lambda = -2(x-1)} \; --\rightarrow \mathbf{(1)}$

- $\frac{\partial F}{\partial y} = 2(y-1) + \lambda(2) = 0$

$\Rightarrow \boxed{\lambda = -(y-1)} \; --\rightarrow \mathbf{(2)}$

Example of Lagrange's Multiplier Method

Solution Continue:

$$F(x,y,z) = \left((x-1)^2 + (y-1)^2 + (z-1)^2\right) + \lambda(x + 2y + 3z - 13)$$

$$\frac{\partial F}{\partial z} = 2(z-1) + \lambda(3) = 0$$

$$\Rightarrow \boxed{\lambda = -\frac{2}{3}(z-1)} \dashrightarrow (3)$$

$$x + 2y + 3z - 13 = 0$$

$$\Rightarrow \boxed{x + 2y + 3z = 13} \dashrightarrow (4)$$

From equations (1), (2) & (3) we get

$$-2(x-1) = -(y-1) = -\frac{2}{3}(z-1)$$

$$\Rightarrow \boxed{2(x-1) = (y-1)} = \frac{2}{3}(z-1)$$

$$2(x-1) = y-1$$

$$\Rightarrow 2x - 2 = y - 1$$

$$\Rightarrow 2x = y + 1$$

$$\Rightarrow \boxed{x = \frac{y+1}{2}} \dashrightarrow (5)$$

Example of Lagrange's Multiplier Method

Solution Continue:

$$x + 2y + 3z = 13 \dashrightarrow (4)$$

$$x = \frac{y+1}{2} \dashrightarrow (5)$$

$$2(x-1) = \boxed{(y-1) = \frac{2}{3}(z-1)}$$

$$(y-1) = \frac{2}{3}(z-1)$$

$$\Rightarrow 3(y-1) = 2(z-1)$$

$$\Rightarrow 3y - 3 = 2z - 2$$

$$\Rightarrow 3y - 1 = 2z$$

$$\Rightarrow \boxed{z = \frac{3y-1}{2}} \dashrightarrow (6)$$

$$\Rightarrow \left(\qquad \right) + 2y + 3\left(\qquad \right) = 13$$

$$\Rightarrow \left(\frac{y+1}{2}\right) + 2y + \left(\frac{9y-3}{2}\right) = 13$$

$$\Rightarrow y + 1 + 4y + 9y - 3 = 26$$

$$\Rightarrow 14y = 28$$

Example of Lagrange's Multiplier Method

Solution Continue:

$$x = \frac{y+1}{2} \dashrightarrow (5)$$

$$z = \frac{3y-1}{2} \dashrightarrow (6)$$

$$14y = 28$$

$$\Rightarrow y = \frac{28}{14}$$

$$\Rightarrow \boxed{y = 2}$$

$$x = \frac{y+1}{2}$$

$$\Rightarrow x = \frac{2+1}{2}$$

$$\Rightarrow \boxed{x = \frac{3}{2}}$$

$$z = \frac{3y-1}{2}$$

$$\Rightarrow z = \frac{3(2)-1}{2}$$

$$\Rightarrow \boxed{z = \frac{5}{2}}$$

Hence, $\boxed{P(x, y, z) = \left(\frac{3}{2}, 2, \frac{5}{2}\right)}$ be the point on the plane $x + 2y + 3z = 13$ closest to the point $Q(1, 1, 1)$.

Example of Lagrange's Multiplier Method

Example 8: Find the shortest and largest distance from the point $(1, 2, -1)$ to the sphere $x^2 + y^2 + z^2 = 24$.

Solution: Let, $P(x, y, z)$ be any point on surface of the sphere $x^2 + y^2 + z^2 = 24$ and $Q(1, 2, -1)$ is the given point.

Hence, distance between P & Q is PQ which is defined as:

$$PQ = \sqrt{(x-1)^2+(y-2)^2+(z+1)^2}$$

$$\Rightarrow PQ^2 = (x-1)^2+(y-2)^2+(z+1)^2 \ \& \ x^2+y^2+z^2 = 24$$

$$\Rightarrow f(x, y, z) = (x-1)^2+(y-2)^2+(z+1)^2$$

$$\& \ \phi(x, y, z) = x^2+y^2+z^2-24 = 0$$

Example of Lagrange's Multiplier Method

Solution Continue:

$f(x,y,z) = (x-1)^2 + (y-2)^2 + (z+1)^2$

$\phi(x,y,z) = x^2 + y^2 + z^2 - 24 = 0$

- Construct the new function

$F(x,y,z)$ as $F(x,y,z) = f(x,y,z) + \lambda\,\phi(x,y,z)$.

$\Rightarrow F(x,y,z) = (\qquad\qquad) + \lambda(\qquad\qquad)$

- Solve the four equations $\dfrac{\partial F}{\partial x} = 0, \dfrac{\partial F}{\partial y} = 0, \dfrac{\partial F}{\partial z} = 0$ & $\phi(x,y,z) = 0$ to find the stationary points.

$$\Rightarrow \frac{\partial F}{\partial x} = 2(x-1) + \lambda(2x) = 0$$

$$\Rightarrow \boxed{\lambda = -\frac{(x-1)}{x}} \dashrightarrow (1)$$

$$\Rightarrow \frac{\partial F}{\partial y} = 2(y-2) + \lambda(2y) = 0$$

$$\Rightarrow \boxed{\lambda = -\frac{(y-2)}{y}} \dashrightarrow (2)$$

Example of Lagrange's Multiplier Method

Solution Continue:

$$F(x,y,z) = \left((x-1)^2 + (y-2)^2 + (z+1)^2\right) + \lambda(x^2 + y^2 + z^2 - 24)$$

$$\frac{\partial F}{\partial z} = 2(z+1) + \lambda(2z) = 0$$

$$\Rightarrow \boxed{\lambda = -\frac{(z+1)}{z}} \dashrightarrow (3)$$

$$x^2 + y^2 + z^2 - 24 = 0$$

$$\Rightarrow \boxed{x^2 + y^2 + z^2 = 24} \dashrightarrow (4)$$

From equations (1), (2) & (3) we get

$$-\frac{(x-1)}{x} = -\frac{(y-2)}{y} = -\frac{(z+1)}{z}$$

$$\Rightarrow \boxed{\frac{(x-1)}{x} = \frac{(y-2)}{y}} = \frac{(z+1)}{z}$$

$$\frac{(x-1)}{x} = \frac{(y-2)}{y}$$

$$\Rightarrow xy - y = xy - 2x$$

$$\Rightarrow -y = -2x$$

$$\Rightarrow \boxed{x = \frac{y}{2}} \dashrightarrow (5)$$

Example of Lagrange's Multiplier Method

Solution Continue:

$$\frac{(x-1)}{x} = \boxed{\frac{(y-2)}{y} = \frac{(z+1)}{z}}$$

$$\frac{(y-2)}{y} = \frac{(z+1)}{z}$$

$$\Rightarrow z(y-2) = y(z+1)$$

$$\Rightarrow yz - 2z = yz + y$$

$$\Rightarrow \boxed{z = -\frac{y}{2}} \dashrightarrow (6)$$

$$x^2 + y^2 + z^2 = 24 \dashrightarrow (4) \qquad x = \frac{y}{2} \dashrightarrow (5)$$

$$\Rightarrow \left(\quad\right)^2 + y^2 + \left(\qquad\right)^2 = 24$$

$$\Rightarrow \frac{y^2}{4} + y^2 + \frac{y^2}{4} = 24$$

$$\Rightarrow y^2 + 4y^2 + y^2 = 96$$

$$\Rightarrow 6y^2 = 96 \quad \Rightarrow y^2 = 16$$

$$\Rightarrow \boxed{y = \pm 4}$$

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Example of Lagrange's Multiplier Method

Solution Continue: $\Rightarrow$ $\boxed{y = \pm 4}$

$$x = \frac{y}{2} \quad - - \rightarrow \mathbf{(5)}$$

$$z = -\frac{y}{2} \quad - - \rightarrow \mathbf{(6)}$$

$$\Rightarrow x = \frac{y}{2}$$

$$\Rightarrow x = \frac{\pm 4}{2}$$

$$\Rightarrow \boxed{x = \pm 2}$$

$$\Rightarrow z = -\frac{y}{2}$$

$$\Rightarrow z = -\frac{\pm 4}{2}$$

$$\Rightarrow \boxed{z = \mp 2}$$

Hence, stationary points are $P(x, y, z) = P(2, 4, -2)$ & $P(-2, -4, 2)$.

Example of Lagrange's Multiplier Method

Solution Continue: Stationary points are $P(x, y, z) = P(2, 4, -2)$ & $P(-2, -4, 2)$.

- $PQ = \sqrt{(x-1)^2 + (y-2)^2 + (z+1)^2}$
- At, point $P(2, 4, -2)$

$$PQ = \sqrt{(2-1)^2 + (4-2)^2 + (-2+1)^2}$$
$$= \sqrt{1+4+1} = \sqrt{6}$$

- At, point $P(-2, -4, 2)$

$$PQ = \sqrt{(-2-1)^2 + (-4-2)^2 + (2+1)^2}$$
$$= \sqrt{9+36+9} = \sqrt{54}$$

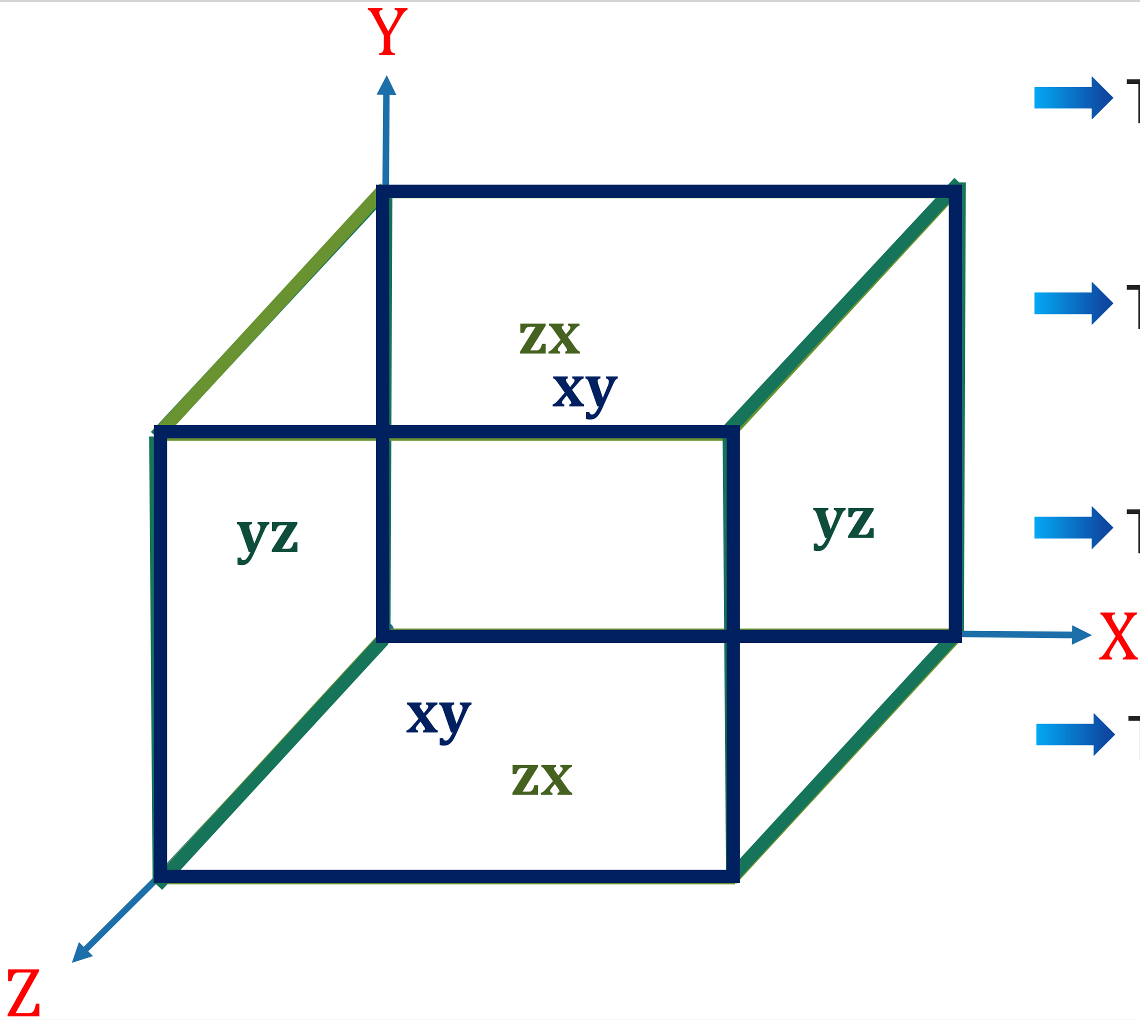
$\Rightarrow \boxed{PQ = \sqrt{6}}$

Which is shortest distance

$\Rightarrow \boxed{PQ = \sqrt{54}}$

Which is largest distance

Lagrange's Multiplier Method



- The area of closed rectangular box is

$$A(x, y, z) = 2xy + 2yz + 2zx$$

- The area of open rectangular box is

$$A(x, y, z) = 2xy + 2yz + zx$$

- The volume of open rectangular box is

$$V(x, y, z) = xyz$$

- The volume of closed rectangular box is

$$V(x, y, z) = xyz$$

Example of Lagrange's Multiplier Method

Example 11: A rectangular box open at the top is to have a given capacity of $64\ \text{cm}^3$. Find the dimension of the box requiring least material for it's construction.

Solution: We have a open rectangular box with capacity $64\ \text{cm}^3$.

$\Rightarrow$ volume $= xyz = 64$

- The area of open rectangular box is $A = 2xy + 2yz + zx$

$$\Rightarrow f(x, y, z) = 2xy + 2yz + zx \ \ \& \ \phi(x, y, z) = xyz - 64 = 0$$

- Construct the new function

$F(x, y, z)$ as $F(x, y, z) = f(x, y, z) + \lambda\, \phi(x, y, z)$.

$$\Rightarrow F(x, y, z) = (\qquad\qquad\qquad) + \lambda(\qquad\qquad)$$

Example of Lagrange's Multiplier Method

Solution Continue: $F(x,y,z) = (2xy + 2yz + zx) + \lambda(xyz - 64)$

- Solve the four equations $\frac{\partial F}{\partial x} = 0, \frac{\partial F}{\partial y} = 0, \frac{\partial F}{\partial z} = 0$ & $\phi(x,y,z) = 0$ to find the stationary points.

- $$\frac{\partial F}{\partial x} = 2y + z + \lambda(yz) = 0$$

$$\Rightarrow \boxed{\lambda = -\frac{(2y+z)}{yz}} \dashrightarrow (\mathbf{1})$$

- $$\frac{\partial F}{\partial y} = 2x + 2z + \lambda(xz) = 0$$

$$\Rightarrow \boxed{\lambda = -\frac{(2x+2z)}{xz}} \dashrightarrow (\mathbf{2})$$

- $$\frac{\partial F}{\partial z} = 2y + x + \lambda(xy) = 0$$

$$\Rightarrow \boxed{\lambda = -\frac{(2y+x)}{xy}} \dashrightarrow (\mathbf{3})$$

- $$xyz - 64 = 0$$

$$\Rightarrow \boxed{xyz = 64} \dashrightarrow (\mathbf{4})$$

Example of Lagrange's Multiplier Method

Solution Continue:

- From equations (1), (2) & (3) we get

$$-\frac{(2y+z)}{yz} = -\frac{(2x+2z)}{xz} = -\frac{(2y+x)}{xy}$$

$$\Rightarrow \boxed{\frac{(2y+z)}{yz}} = \boxed{\frac{(2x+2z)}{xz}} = \boxed{\frac{(2y+x)}{xy}}$$

$$\Rightarrow xz(2y+z) = yz(2x+2z)$$

$$\Rightarrow \cancel{2xyz} + xz^2 = \cancel{2xyz} + 2yz^2$$

$$\Rightarrow x\cancel{z^2} = 2y\cancel{z^2}$$

$$\Rightarrow \boxed{x = 2y}$$

$$\Rightarrow xy(2x+2z) = xz(2y+x)$$

$$\Rightarrow 2x^2y + \cancel{2xyz} = \cancel{2xyz} + x^2z$$

$$\Rightarrow 2\cancel{x^2}y = \cancel{x^2}z$$

$$\Rightarrow \boxed{z = 2y}$$

- From equation (4) $xyz = 64$

$$\Rightarrow (\quad)y(\quad) = 64$$

$$\Rightarrow 4y^3 = 64 \Rightarrow y^3 = 16$$

$$\Rightarrow \boxed{y = 2\sqrt[3]{2}}$$

Example of Lagrange's Multiplier Method

Solution Continue: $y = 2\sqrt[3]{2}$

$\rightarrow x = 2y$

$\Rightarrow \boxed{x = 4\sqrt[3]{2}}$

$\rightarrow z = 2y$

$\Rightarrow \boxed{z = 4\sqrt[3]{2}}$

$\Rightarrow$ Dimensions $= \boxed{(x, y, z) = \left(4\sqrt[3]{2}, 2\sqrt[3]{2}, 4\sqrt[3]{2}\right)}.$

$\rightarrow$ **Note:** The possible answers are depends on formula of surface area.

$\rightarrow$ If $A = 2xy + yz + 2zx$ then $(x, y, z) = \left(2\sqrt[3]{2}, 4\sqrt[3]{2}, 4\sqrt[3]{2}\right)$.

$\rightarrow$ If $A = xy + 2yz + 2zx$ then $(x, y, z) = \left(4\sqrt[3]{2}, 4\sqrt[3]{2}, 2\sqrt[3]{2}\right)$.

Method:9 $\rightarrow$ Gradient

Gradient of a Scalar Function

- A function whose codomain is scalars is called a scalar function.

  Example: $f(x, y, z) = 2x^2y + 4z$ is a scalar function.

- Del operator $= \nabla(\text{nabla}) = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) = \frac{\partial}{\partial x}\,\hat{\imath} + \frac{\partial}{\partial y}\,\hat{\jmath} + \frac{\partial}{\partial z}\,\hat{k}.$

- The gradient of a scalar function $f(x, y, z)$ is denoted by grad $f/\ \nabla f$ defined as

$$\text{grad } f = \nabla f = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right) = \frac{\partial f}{\partial x}\,\hat{\imath} + \frac{\partial f}{\partial y}\,\hat{\jmath} + \frac{\partial f}{\partial z}\,\hat{k}$$

- The gradient of scalar function is a vector.

Example of Gradient of a Scalar Function

Example 1: Find $\nabla\phi$ at $(1,-2,-1)$, if $\phi = 3x^2y - y^3z^2$.

Solution: Here, $\phi = 3x^2y - y^3z^2$.

$$\Rightarrow \nabla\phi = \left(\frac{\partial\phi}{\partial x}, \frac{\partial\phi}{\partial y}, \frac{\partial\phi}{\partial z}\right)$$

$$= (6xy,\ 3x^2 - 3y^2z^2,\ -2y^3z\ )$$

$$\Rightarrow \nabla\phi_{(1,-2,-1)} = \left(6(1)(-2), 3(1)^2 - 3(-2)^2(-1)^2, -2(-2)^3(-1)\right)$$

$$= (-12, 3 - 12, -16)$$

$$\Rightarrow \boxed{\nabla\phi_{(1,-2,-1)} = (-12,-9,-16)}$$

Example of Gradient of a Scalar Function

Example 5: Evaluate ∇e^{r^2}, where $r^2 = x^2 + y^2 + z^2$.

Solution: Let, $f = e^{r^2} = e^{x^2+y^2+z^2}$.

$$\Rightarrow \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right)$$

$$= \left(e^{x^2+y^2+z^2} \cdot (2x), e^{x^2+y^2+z^2} \cdot (2y),\ e^{x^2+y^2+z^2} \cdot (2z)\right)$$

$$= 2e^{x^2+y^2+z^2}(x, y, z)$$

$$\Rightarrow \boxed{\nabla f = 2e^{r^2}(x, y, z)}$$

Method:10 → Directional Derivative

Directional Derivative

- The directional derivative of $f(x, y)$ at (a, b) in the direction $\bar{u} = (u_1, u_2)$ is

$$\boxed{D_{\bar{u}} f = (\operatorname{grad} f)_{(a,b)} \cdot \frac{\bar{u}}{|\bar{u}|}}$$

$$= \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)_{(a,b)} \cdot \frac{(u_1, u_2)}{|\bar{u}|}$$

$$= \left(f_{x(a,b)}, f_{y(a,b)}\right) \cdot \left(\frac{u_1}{|\bar{u}|}, \frac{u_2}{|\bar{u}|}\right)$$

$$\Rightarrow \boxed{D_{\bar{u}}\ f(a, b) = f_x(a, b)\frac{u_1}{|\bar{u}|} + f_y(a, b)\frac{u_2}{|\bar{u}|}}.$$

Directional Derivative

- The directional derivative of $f(x, y, z)$ at (x_0, y_0, z_0) in the direction of $\bar{u} = (u_1, u_2, u_3)$ is

$$\boxed{D_{\bar{u}} f(x_0, y_0, z_0) = f_x(x_0, y_0, z_0)\frac{u_1}{|\bar{u}|} + f_y(x_0, y_0, z_0)\frac{u_2}{|\bar{u}|} + f_z(x_0, y_0, z_0)\frac{u_3}{|\bar{u}|}}$$

- The directional derivative of $f(x, y)$ in direction of unit vector $\bar{u} = (u_1, u_2)$ which makes an angle α with positive x-axis and at point (x_0, y_0) is

$$\boxed{D_{\alpha} f(x_0, y_0) = \cos\alpha \, f_x(x_0, y_0) + \sin\alpha \, f_y(x_0, y_0)}$$

Example of Directional Derivative

Example 1: Find directional derivative of $f(x, y) = xe^y + \cos xy$ at the point $(2, 0)$ in the direction $\bar{A} = 3\hat{i} - 4\hat{j}$.

Solution: Here, $f(x, y) = xe^y + \cos xy$,

Point $\bar{x} = (a, b) = (2, 0)$ & Direction $\bar{A} = (3, -4)$.

$$D_{\bar{x}} f = (\nabla f)_{(a,b)} \cdot \frac{\bar{A}}{|\bar{A}|} \dashrightarrow (1)$$

Now, $\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right) = (e^y - y \sin xy\,,\; xe^y - x \sin xy)$

$$\Rightarrow \nabla f_{(2,0)} = \left(e^0 - (0)\sin(2 \cdot 0), (2)e^{(0)} - (2)\sin(2 \cdot 0)\right)$$

$$\Rightarrow \boxed{\nabla f_{(2,0)} = (1, 2)}$$

Example of Directional Derivative

Example 1: Find directional derivative of $f(x, y) = xe^y + \cos xy$ at the point $(2, 0)$ in the direction $\bar{A} = 3\hat{\imath} - 4\hat{\jmath}$.

Solution Continue:

Here, $\bar{A} = 3\hat{\imath} - 4\hat{\jmath} = (3, -4)$. $\Rightarrow |\bar{A}| = \sqrt{3^2 + (-4)^2} = \sqrt{25} = 5$.

$$\Rightarrow \frac{\bar{A}}{|\bar{A}|} = \frac{1}{5}(3, -4) = \left(\frac{3}{5}, \frac{-4}{5}\right)$$

$$D_{\bar{x}} f = (\nabla f)_{(a,b)} \cdot \frac{\bar{A}}{|\bar{A}|} = (1, 2) \cdot \left(\frac{3}{5}, \frac{-4}{5}\right) = 1\left(\frac{3}{5}\right) + 2\left(\frac{-4}{5}\right) = -\frac{5}{5}$$

$$\Rightarrow \boxed{D_{\bar{A}} f = -1}$$

Example of Directional Derivative

Example 3: Define gradient of a function. Use it to find directional derivative of $f(x,y,z) = x^3 - xy^2 - z$ at $P(1,1,0)$ in the direction $\bar{a} = 2\hat{\imath} - 3\hat{\jmath} + 6\hat{k}$.

Solution: The gradient of a function $f(x,y,z)$ is denoted by $\text{grad } f / \nabla f$ defined as

$$\text{grad } f = \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right).$$

Here, $f = x^3 - xy^2 - z$, Point $= \bar{x} = (1,1,0)$ & Direction $\bar{a} = (2,-3,6)$.

$$\Rightarrow \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right) = (3x^2 - y^2, -2xy, -1)$$

$$\Rightarrow \nabla f_{(1,1,0)} = (3(1)^2 - (1)^2, -2(1)(1), -1)$$

$$\Rightarrow \boxed{\nabla f_{(1,1,0)} = (2,-2,-1)}$$

Example of Directional Derivative

Example 3: Define gradient of a function. Use it to find directional derivative of $f(x,y,z) = x^3 - xy^2 - z$ at $P(1,1,0)$ in the direction $\bar{a} = 2\hat{\imath} - 3\hat{\jmath} + 6\hat{k}$.

Solution Continue: Here, $\bar{a} = 2\hat{\imath} - 3\hat{\jmath} + 6\hat{k} = (2,-3,6)$.

$$\Rightarrow |\bar{a}| = \sqrt{2^2 + (-3)^2 + 6^2} = \sqrt{4+9+36} = \sqrt{49} = 7$$

$$\Rightarrow \frac{\bar{a}}{|\bar{a}|} = \frac{1}{7}(2,-3,6) = \left(\frac{2}{7}, -\frac{3}{7}, \frac{6}{7}\right)$$

$$D_{\bar{x}} f = (\nabla f)_{(1,1,0)} \cdot \frac{\bar{a}}{|\bar{a}|} = (2,-2,-1) \cdot \left(\frac{2}{7}, -\frac{3}{7}, \frac{6}{7}\right)$$

$$= 2\left(\frac{2}{7}\right) - 2\left(-\frac{3}{7}\right) - 1\left(\frac{6}{7}\right) = \frac{4}{7} + \frac{6}{7} - \frac{6}{7} = \frac{4}{7}$$

$$\Rightarrow \boxed{D_{\bar{x}} f = \frac{4}{7}}$$

Example of Directional Derivative

Example 10: Find the directional derivative of $x^2y^2z^2$ at $(1,1,-1)$ along a direction equally inclined with co-ordinate axis.

Solution: Here, $f = x^2y^2z^2$, Point $\overline{x} = (1,1,-1)$ & Direction $\overline{a} = (1,1,1)$.

$$\Rightarrow \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right) = (2xy^2z^2,\ 2yx^2z^2,\ 2zx^2y^2)$$

$$\Rightarrow \nabla f_{(1,1,-1)} = (2(1)(1)^2(-1)^2, 2(1)(1)^2(-1)^2, 2(-1)(1)^2(1)^2)$$

$$\Rightarrow \nabla f_{(1,1,-1)} = (2,2,-2).$$

➡ Here, $\bar{a} = (1,1,1) \Rightarrow |\bar{a}| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$.

$$\Rightarrow \frac{\bar{a}}{|\bar{a}|} = \frac{1}{\sqrt{3}}(1,1,1) = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

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Example of Directional Derivative

Example 10: Find the directional derivative of $x^2y^2z^2$ at $(1, 1, -1)$ along a direction equally inclined with co-ordinate axis.

Solution Continue:

$$\Rightarrow D_{\bar{x}} f = (\nabla f)_{(1,1,-1)} \cdot \frac{\bar{a}}{|\bar{a}|} = (2, 2, -2) \cdot \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

$$= 2\left(\frac{1}{\sqrt{3}}\right) + 2\left(\frac{1}{\sqrt{3}}\right) - 2\left(\frac{1}{\sqrt{3}}\right) = \frac{2}{\sqrt{3}}$$

$$\Rightarrow \boxed{D_{\bar{A}} f = \frac{2}{\sqrt{3}}}$$

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Example of Directional Derivative

Example 12: Find the directional derivative $D_u f(x,y)$ if $f(x,y) = x^3 - 3xy + 4y^2$ and u is the unit vector given by angle $\theta = \frac{\pi}{6}$.What is $D_u f(1,2)$?

Solution:

- The directional derivative of $f(x,y)$ in direction of unit vector $u = (u_1, u_2)$ which makes an angle θ with positive x-axis and at point $(1,2)$ is

$$D_u f(1,2) = \cos\theta \; f_x(1,2) + \sin\theta \; f_y(1,2).$$

- Here, $f(x,y) = x^3 - 3xy + 4y^2$
- $f_x = 3x^2 - 3y$

$\Rightarrow f_x(1,2) = 3(1)^2 - 3(2) = -3$

- $f_y = -3x + 8y$

$\Rightarrow f_y(1,2) = -3(1) + 8(2) = 13$

Example of Directional Derivative

Example 12: Find the directional derivative $D_u f(x,y)$ if $f(x,y) = x^3 - 3xy + 4y^2$ and u is the unit vector given by angle $\theta = \frac{\pi}{6}$. What is $D_u f(1,2)$?

Solution Continue:

$$D_u f(1,2) = \cos\frac{\pi}{6} f_x(1,2) + \sin\frac{\pi}{6} f_y(1,2)$$

$$= \left(\frac{\sqrt{3}}{2}\right)(-3) + \left(\frac{1}{2}\right)(13)$$

$$\Rightarrow \boxed{D_u f(1,2) = \frac{13 - 3\sqrt{3}}{2}}$$

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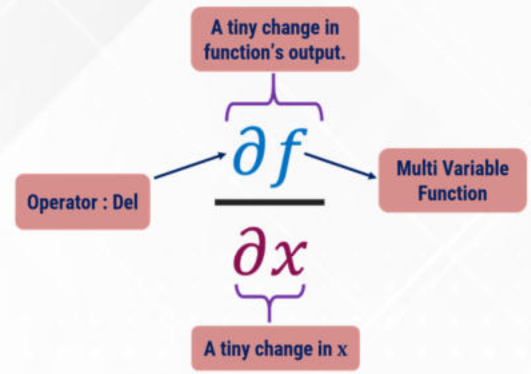

THANK YOU

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